A Novel Cross-Scale Octave Network for Hyperspectral and Multispectral Image Fusion

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***Abstract*— Recently, deep convolutional neural network-based low-resolution hyperspectral image (LR-HSI) and high-resolution multispectral image (HR-MSI) fusion methods have achieved significant performance improvement. However, the rich spatial and spectral information in HSIs is not fully explored. In this article, we propose a novel cross-scale octave network (CSONet) for hyperspectral and multispectral image fusion. Specifically, we adopt a progressive image fusion structure to effectively extract the spatial and spectral information of HR-MSI at multiple resolutions, thereby efficiently complementing LR-HSI's information. In addition, the proposed cross-scale octave convolution module can extract rich multiscale spatial feature information and concentrate on more important spatial–spectral features at different scales with the multiscale spatial–spectral attention mechanism. Finally, a multisupervised loss function is used to improve the gradient propagation and enhance the representation ability of the network. Ablation analysis on the benchmark datasets shows the effectiveness of each component in the proposed method. Extensive experimental results on different hyperspectral images demonstrate that the proposed CSONet can achieve superior results and strong generalization ability in comparison with some state-of-the-art LR-HSI and HR-MSI fusion methods.**

***Index Terms*— Attention mechanism, convolutional neural network (CNN), cross-scale octave convolution, hyperspectral and multispectral image fusion.**

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I. Introduction

HYPERSPECTRAL images have been widely used in many different fields because of their rich spatial and spectral information, such as target recognition [1], classification [2], and land exploration [3]. However, due to physical limitations, the information provided by a single sensor is incomplete. The resolution of remote sensing images obtained from a single sensor can only be a compromise between spatial and spectral resolution [4]. For example, low-resolution hyperspectral images (LR-HSIs) usually have a high spectral resolution, but their spatial resolution is coarse; high-resolution multispectral images (HR-MSIs) have a higher spatial resolution, but the spectral resolution is relatively low. Therefore, how to obtain a hyperspectral image with the high spatial resolution has become an interesting research topic. Hyperspectral and multispectral image fusion is an important approach to improve the spatial resolution of the observed LR-HSI. The objective of hyperspectral and multispectral image fusion is to integrate the fine spatial texture information contained in HR-MSI and the rich spectral information contained in LR-HSI to obtain a fused image with high spatial and spectral resolution. With the increasing demand for high-resolution remote sensing data, many remote sensing image fusion methods (including pan-sharpening and HSI super-resolution) have been proposed in recent years, such as component substitution (CS)-based methods [5], [6], multiresolution analysis (MRA)-based methods [7], model-based approaches [8], and deep learning-based methods [9].

In the CS-based methods, spectral conversion is performed on the MS image so that one or all of the components are replaced by the PAN image to match the histogram. Commonly used methods based on component replacement include the intensity-hue-saturation (IHS) method [10] and the principal component analysis (PCA) method [11]. The advantage of this method is that the geometric structure is well preserved, thereby obtaining an output image with fine spatial details. On the other hand, the component replacement-based methods carry additional spectral information and cause spectral distortion. The methods based on MRA are expected to infer the missing spatial details of MS images from PAN images and inject them into MS images. In order to improve the effectiveness of the detail injection, wavelet-based methods are proposed to perform image fusion. The model-based method reconstructs the hyperspectral images with high spatial

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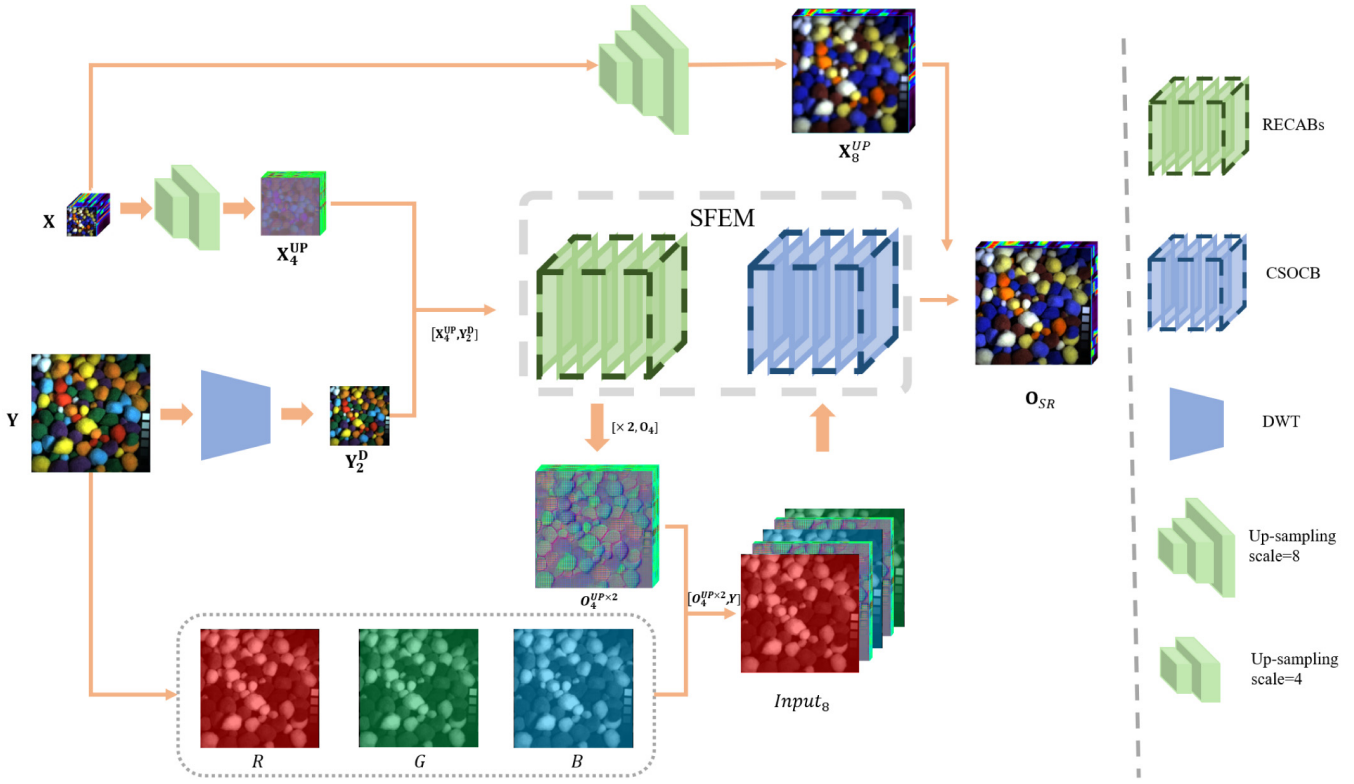


Fig. 1. Overall structure of the proposed CSONet.

resolution (HR-HSI) by assuming the relationship between the LR-HSI and the HR-MSI for model construction, and transforms the image reconstruction problem into a problem of minimizing the objective function. Although the model-based method has a tradeoff between spatial resolution enhancement and spectral preservation, it is still ill-posed and has high computational complexity.

In recent years, with the rapid development of deep learning techniques, many natural image super-resolution methods have been proposed to learn the mapping relationship between low-resolution images and their corresponding high-resolution images [12], [13], [14], [15], [16], [17], [18], [19]. Chao et al. [20] proposed the first image super-resolution method (SRCNN) based on a convolutional neural network (CNN). Lately, many natural super-resolution methods based on residual learning, dense connections [21], and GAN [22] are proposed, achieving significant performance improvement. Inspired by the great success of CNN in the field of natural image super-resolution, researchers have proposed many HSI super-resolution methods based on deep learning to fuse the observed HR-MSI and LR-HSI to obtain an HR-HSI. Masi et al. [23] proposed a CNN-Based pan-sharpening (PNN) for remote sensing image fusion, which achieved better results than traditional methods. Taking into account that a network with a deep structure may cause performance degradation, He et al. [24] introduced residual learning into the network. Dian et al. [25] proposed a deep HSI sharpening (DHSIS) method, which directly learns image prior information through residual learning and integrates the learned deep prior knowledge into an LR-HSI and HR-MSI fusion framework. Although these methods have achieved good results, there are still problems in the preservation of spatial and spectral information. Jingxiang et al. [26] used two branched deep structures to extract spatial and spectral information, respectively. Yuan et al. [15] proposed a multiscale and multidepth CNN (MSDCNN), which introduced multiscale feature extraction and residual learning into the CNN. Hu et al. [27] adopted MRA and attention mechanisms to efficiently extract spatial and spectral feature information. Zheng et al. [28] proposed a high computational efficiency network to simultaneously explore spatial and spectral information. Zhu et al. [29] proposed a progressive zero-centric residual network (PZRes-Net) to address the cross-modality distribution problem of the spatial and spectral information. In addition to the fusion-based methods, there are other promising methods to improve the quality of HS images, e.g., single RGB image-based [30], [31] and HS image spatial reconstruction-based [32].

To further improve the performance of LR-HSI and HR-MSI fusion, we propose a novel cross-scale octave network (CSONet), as shown in Fig. 1. Considering that LR-HSI contains most of the spectral information and some spatial information, and HR-MSI has a large number of high-resolution spatial details and few spectral information, the proposed CSONet adopts a two-branch architecture to learn the rich spatial–spectral features. Specifically, one of the branches is in charge of preserving the spectral information by upsampling the LR-HSI in the global residual connection, and the other one is mainly responsible for capturing the spatial details and some spectral information. In the proposed cross-scale octave

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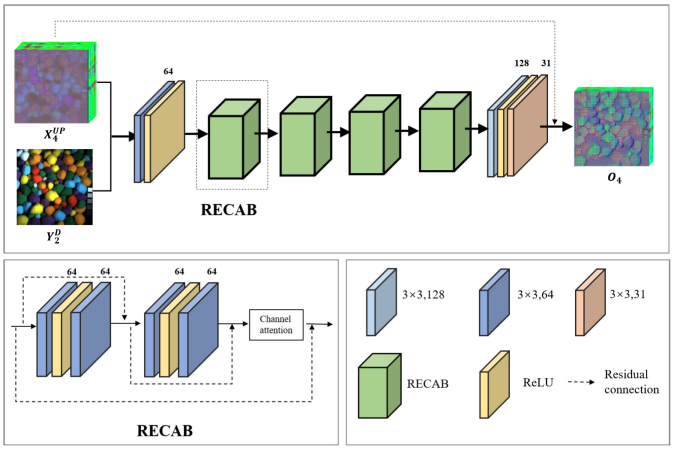


Fig. 2. Network structure of RECABs.

block, it enables the model to learn rich multiscale feature information and improves the feature propagation among different scales and layers, which is beneficial to enhance the representation of the model. In addition, benefiting from the proposed multiscale spatial–spectral attention mechanism (MSSSA), our model can learn the discriminative spectral and spatial features and enhance the representation ability of the model. Finally, we adopt a multisupervised learning strategy to improve the flow of gradient information and alleviate exploding/vanishing gradients.

The main contributions of this article are given as follows.

1) This article proposes a CSONet for LR-HSI and HR-MSI fusion. The proposed network can learn rich multiscale contextual information and improve the performance of image fusion.
2) We introduce a novel cross-scale octave block, which can fully exploit spatial–spectral feature information across different scales. Moreover, the proposed MSSSA can efficiently capture important spectral and spatial features to adaptively recalibrate the feature representations of the model. A multisupervised loss function is designed to improve the flow of information. Extensive experimental results show that our proposed method can achieve better results in comparison with the state-of-the-art HSI SR methods.

The remainder of this article is organized as follows. Section II presents the related work. Section III describes the proposed CSONet in detail. A series of experiments and discussions are provided in Section IV. Conclusions are drawn in Section V.

II. Related Works

In this section, we will review three previous works closely related to our proposed network, i.e., CNN-based multispectral and hyperspectral image fusion [13], [33], [34], multiscale network, and attention mechanism.

A. CNN-Based Multispectral and Hyperspectral Image Fusion

Let $\mathrm{LR}_{\mathrm{hs}} \in \mathbb{R}^{w\times h\times C}$ and $\mathrm{HR}_{\mathrm{ms}} \in \mathbb{R}^{W\times H\times c}$ denote an LR-HSI and an HR-MSI, respectively, where w, h, C, and W, H, c are the width, the length, and the number of channels of LR-HSI and HR-MSI, respectively. The existing CNN-based multispectral and hyperspectral image fusion methods can be roughly divided into two types: the single-branch fusion network and the multibranch fusion network. In a single-branch network structure, LR-HSI is first upsampled to obtain $\mathrm{LR_up}_{\mathrm{hs}} \in \mathbb{R}^{W\times H\times C}$ that has the same spatial size as $\mathrm{HR}_{\mathrm{ms}} \in \mathbb{R}^{W\times H\times c}$. Then, $\mathrm{HR}_{\mathrm{ms}}$ and $\mathrm{LR_up}_{\mathrm{hs}}$ are concatenated as the input of the network to learn the target hyperspectral image $\mathrm{SR}_{\mathrm{hs}} \in \mathbb{R}^{W\times H\times C}$ with high spatial resolution. In a multibranch fusion network, $\mathrm{HR}_{\mathrm{ms}}$ and $\mathrm{LR}_{\mathrm{hs}}$ are directly inputted into different branches to extract the spatial and spectral information, respectively. Subsequently, the output feature maps of the LR-HSI branch are upsampled to the same size as the target image, and then, the output feature maps of the two branches are concatenated along the channel dimension to further obtain the desired super-resolution hyperspectral image (SR-HSI) $\mathrm{SR}_{\mathrm{hs}}$ Specifically, Hu et al. [27] adopt a multibranch network architecture for LR-HSI/HR-MSI fusion, which can achieve better image fusion performance in terms of spatial and spectral aspects. Although these methods obtained significant image fusion results, there is still room to improve spatial reconstruction and spectral preservation.

B. Multiscale Architecture

Multiscale networks have been widely used in the field of computer vision due to their strong representation ability. Chen et al. [35] proposed a multiscale convolutional network to fully exploit the multiscale information through convolution. Weng and Zhu [36] proposed a U-shaped network structure (i.e., U-Net), which divides a feature map into different scales by using expansion and shrinkage paths. Subsequently, to further improve the representation of the network, a novel multiscale network called Oct-Conv was proposed in [37]. Oct-Conv splits the network into two parallel branches to learn different spatial-frequency features. Moreover, different branches can exchange information through cross-scale skip connections; this operation can not only capture more contextual information but also reduce computational complexity. Inspired by Oct-Conv, a series of multiscale convolutions are proposed to enhance the feature representation ability of the network [38]. In this article, we introduce a cross-scale octave convolution structure to fully extract multiresolution feature information in the LR-HSI/HR-MSI fusion task and improve the quality of image reconstruction.

C. Attention Mechanism

In recent years, attention mechanism has made great progress in the field of computer vision [39], [40], [41], [42], [43], [44], [45]. It enables to selectively focus on more important features and suppress nonimportant features, instead of treating all feature contents equally. Wang et al. [46] proposed a CNN-based effective channel attention network (ECA-Net) with few parameters and high performance that does not require dimensionality reduction. Thus, we introduce a multiscale spatial–spectral attention module to improve the representation of spatial and spectral features, which not only

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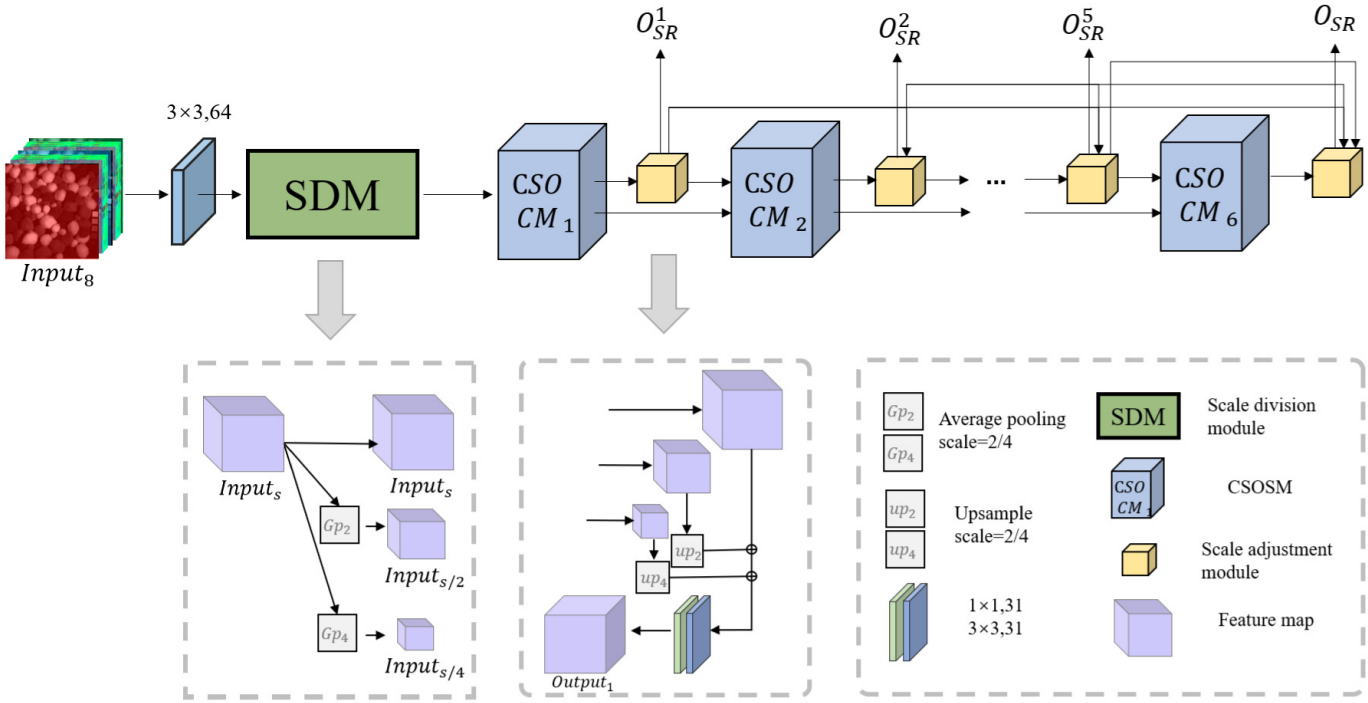


Fig. 3. Flowchart of CSOCB.

enables our method to capture the multiscale spatial–spectral attention information but also further enhances the spectral preservation without dimension reduction. The proposed multiscale spatial–spectral attention module will be introduced in detail in Section III.

III. Proposed CSONet

The flowchart of the network architecture is shown in Fig. 1.

A. Network Architecture

As shown in Fig. 1, the proposed CSONet mainly consists of two parts: the spatial–spectral feature extraction module (SFEM) and the global connection part, which aim to better fuse the key spatial and spectral information from LR-HSI and HR-MSI.

Considering that LR-HSI and HR-MSI contain spatial details and spectral information, respectively, we concatenate LR-HSI and HR-MSI together and then feed them into the network to learn the mapping function between the input LR-HSI/HR-MSI images and the output hyperspectral image. Specifically, the upsampled LR-HSI X $(\times 4, X_4^{\mathrm{UP}})$ and the downsampled HR-MSI Y $(\times 2, Y_2^D)$ are first concatenated to form a cube and then used as the input of several residual blocks with an efficient channel attention mechanism (RECABs) to extract the spatial and spectral feature information at $\times 4$ scale factor. Subsequently, the output features of the above block are upsampled by 2 factor and then concatenated with HR-MSI Y to further learn the rich spatial–spectral feature contents at $\times 8$ scale factor from the proposed cross-scale octave convolution block (CSOCB). Finally, the output features of the spatial–spectral feature extraction part are added to the upsampled LR-HSI to obtain the desired SR-HSI as

$$O_{\mathrm{SR}} = f_\theta(X, Y) + X_8^{\mathrm{UP}} \tag{1}$$

where O_{SR} denotes the desired high-resolution hyperspectral image and θ is the training parameters of the proposed network.

B. Spatial–Spectral Feature Extraction Module

As shown in Fig. 1, the proposed SFEM mainly includes RECABs and a CSOCB.

In the beginning, the combined feature maps $([X_4^{\mathrm{UP}}, Y_2^D])$ are first fed into a convolutional layer followed by an ReLU activation layer and then sent into a series of RECAB, as shown in Fig. 2. The first convolutional layer is to extract the shallow features, and a series of RECABs is used to further extract the spatial–spectral features

$$O_4 = f_4\left(\left[X_4^{\mathrm{UP}}, Y_2^D\right]\right) \tag{2}$$

where O_4 and f_4 denote the output feature and the training function of the above network at $\times 4$ scale factor, respectively, and $[\cdot, \cdot]$ is the concatenation operation. Subsequently, the upsampled feature O_4 is combined with the HR-MSI Y as the input of a series of CSOCB to further learn the spatial–spectral information and inject them into the target high-resolution hyperspectral image.

1) Cross-Scale Octave Convolution Block: The structure of CSOCB is shown in Fig. 3. The input feature $\mathrm{Input}_8(i.e., [O_4^{\mathrm{UP}\times 2}, Y])$ is first fed into a 3×3 convolution layer to adjust the feature channels (Input_s). Then, in order

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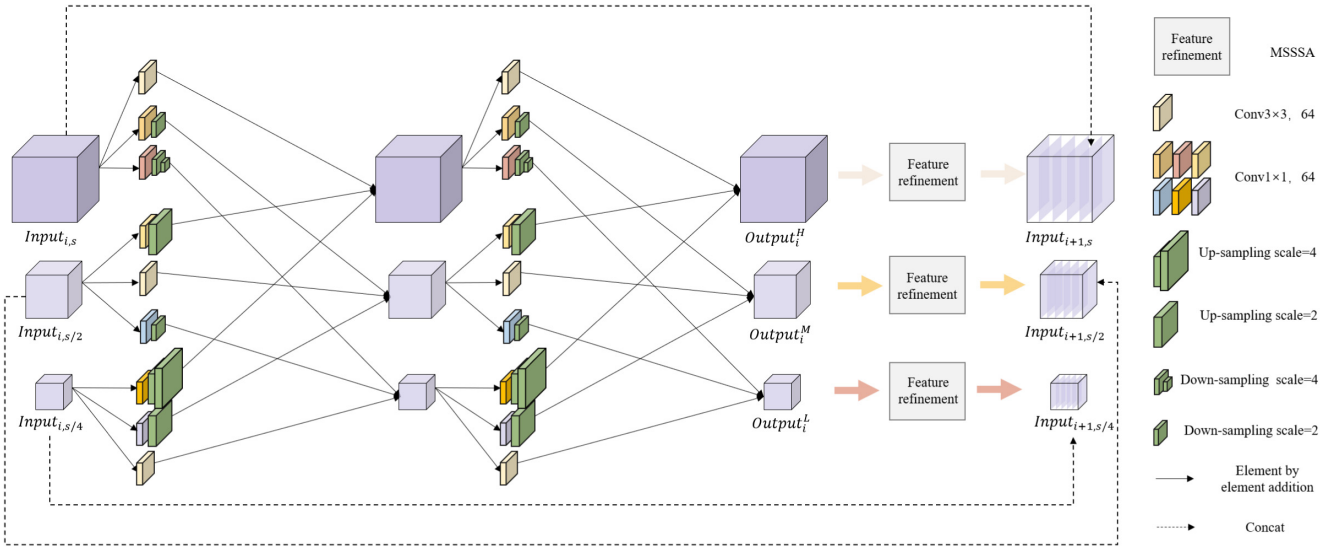


Fig. 4. Network structure of the ith CSOCM.

to obtain multiscale feature maps $\text{Input}_{s/2}$ and $\text{Input}_{s/4}$, the global average pooling operation is performed on Input_s

$$\text{Input}_{s/2} = Gp_2(\text{Input}_s) \tag{3}$$

$$\text{Input}_{s/4} = Gp_4(\text{Input}_s) \tag{4}$$

where $Gp_2(\cdot)$ and $Gp_4(\cdot)$ denote downsampling by a factor 2 and 4 via global average pooling, respectively. Three different scale feature maps, Input_s, $\text{Input}_{s/2}$, and $\text{Input}_{s/4}$, are fed into a series of CSOCMs for multiscale feature extraction. It is worth noting that each CSOCM has three different scales of input and output. Specifically, the input and output of the ith CSOCM can be represented as follows:

$$\begin{matrix}\text{Output}_{i,s}\\ \text{Output}_{i,s/2}\\ \text{Output}_{i,s/4}\end{matrix} = \text{CS}_{i-1}\left(\text{CS}_{i-2}, \ldots, \left(\text{CS}_1\begin{pmatrix}\text{Input}_s\\ \text{Input}_{s/2}\\ \text{Input}_{s/4}\end{pmatrix}\right)\right) \tag{5}$$

where CS_{i-1} represents the $(i-1)$th CSOCM. Three different scale outputs serve two purposes: the first one is to be used as the input of the $(i+1)$th CSOCM, and the second is to upsample $\text{Output}_{i,s/2}$ and $\text{Output}_{i,s/4}$ to the same scale as $\text{Output}_{i,s}$ for these three feature maps to be added to generate the intermediate output feature Output_i. Considering the complementary information between different CSOCMs, we adopt a hierarchical feature fusion mechanism to learn more spatial–spectral feature information. The corresponding output of the ith CSOCM is represented as follows:

$$O_{\text{SR}}^i = (W_2(W_1([\text{Output}_i, \text{Output}_{i-1}, \text{Output}_1])) + X_8^{\text{UP}} \tag{6}$$

where W_1 and W_2 are a 1×1 convolution layer and a 3×3 convolution layer, respectively, which aim to adjust the channel dimensions of output feature and refine spatial and spectral information, respectively.

In addition, to achieve the information exchange among different branches, each branch can be further divided into intrascale and interscale communications. The structure of the ith CSOCM is shown in Fig. 4, For intrascale communication, we directly use a convolutional layer to obtain features. For interscale communication, we first resize the other two features to match the feature size of the current branch and then add them to generate the corresponding output feature. The corresponding implementation is shown as

$$\begin{bmatrix}\text{Output}_i^H\\ \text{Output}_i^M\\ \text{Output}_i^L\end{bmatrix} = \begin{bmatrix} W_{(HH)} & \uparrow_{\text{up}} \cdot W_{(MH)} & \uparrow_{\text{up}} \cdot W_{(LH)} \\ W_{(HM)} \cdot \downarrow_{\text{avg}} & W_{(MM)} & \uparrow_{\text{up}} \cdot W_{(LM)} \\ W_{(HL)} \cdot \downarrow_{\text{avg}} & W_{(ML)} \cdot \downarrow_{\text{avg}} & W_{(LL)} \end{bmatrix} \cdot \begin{bmatrix} W_{(HH)} & \uparrow_{\text{up}} W_{(MH)} & \uparrow_{\text{up}} W_{(LH)} \\ W_{(HM)} \cdot \downarrow_{\text{avg}} & W_{(MM)} & \uparrow_{\text{up}} \cdot W_{(LM)} \\ W_{(HL)} \cdot \downarrow_{\text{avg}} & W_{(ML)} \cdot \downarrow_{\text{avg}} & W_{(LL)} \end{bmatrix} \begin{bmatrix}\text{Input}_{i,s}\\ \text{Input}_{i,s/2}\\ \text{Input}_{i,s/4}\end{bmatrix} \tag{7}$$

where $W_{(\cdot)}$ represents different convolution operations, and $\uparrow_{\text{up}}$ and $\downarrow_{\text{avg}}$ represent upsampling and downsampling operations, respectively.

It is worth noting that the weight-sharing strategy is adopted in intrascale learning, *i.e.*, $W_{(HH)} = W_{(MM)} = W_{(LL)}$. By sharing the weight parameters, the complexity of the model can be effectively reduced while performance stability being maintained.

2) Multiscale Spatial–Spectral Attention Mechanism: Recently, attention mechanisms have been proposed to simulate the human vision system by selectively focusing on more important features and suppressing less important features. Yuan et al. [27] introduced channel attention and spatial attention into the hyperspectral image super-resolution to recalibrate more noteworthy spectral and spatial features. Although the above method can enhance the model performance, they ignore the importance of spatial–spectral attention representations at different scales. Moreover, in channel attention, dimension reduction may lead to the loss of spectral information, thereby affecting the representation ability of spectral bands.

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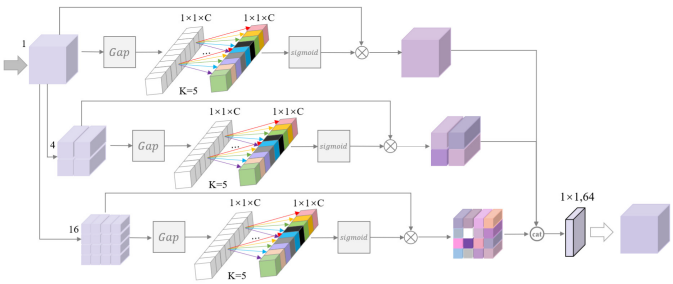


Fig. 5. Flowchart of MSSSA.

In this article, we introduce an efficient MSSSA to further improve the performance of hyperspectral image and multispectral image fusion. The structure of MSSSA is shown in Fig. 5, which consists of three parallel branches. Specifically, the input feature is first divided into one, four, and 16 small feature maps. Then, each small feature is recalibrated in a channelwise manner without dimensionality reduction. Finally, the output features of three different branches are concatenated to learn more informative information and enhance feature representation. It is worth noting that the proposed MSSSA can be degraded to an ECA module when the input feature is not divided.

C. Loss Function

In order to fully utilize feature information and reduce the discrepancy between the target image and the estimated image, a multisupervised loss function is designed to train the network. First, we take the last output of CSONet as the estimated image and adopt the $L1$ loss function to evaluate the similarity between the target image and the estimated image

$$\ell_1(Z, O_{\mathrm{SR}}) = \|Z - O_{\mathrm{SR}}\|_1 \tag{8}$$

where Z and O_{SR} denote the ground-truth (GT) high-resolution hyperspectral image and the reconstructed hyperspectral super-resolution image, respectively.

To evaluate the spectral fidelity, we adopt the spectral angle mapper (SAM) to minimize the spectral difference between the two images

$$\ell_2(Z, O_{\mathrm{SR}}) = \frac{1}{HW}\sum_{i=1}^{H}\sum_{j=1}^{W} \times \arccos\left(\frac{Z(i,j), O_{\mathrm{SR}}(i,j)}{\|Z(i,j)\|_2\|O_{\mathrm{SR}}(i,j)\|_2 + c}\right) \tag{9}$$

where H and W are the height and width of the feature map, respectively, $Z(i,j)$ and $O_{\mathrm{SR}}(i,j)$ are the spectral vector at position (i,j) of the images, and c is a positive constant.

To further improve the reconstruction quality of the model, the output feature of each CSOCM is added to the upsampled LR-HSI to obtain the intermediate reconstructed image O_{SR}^k. Then, the intermediate output is combined with GT to calculate the spatial and spectral loss

$$\ell_3\left(Z, O_{\mathrm{SR}}^k\right) = \sum_{k=1}^{n-1}\left\|Z - O_{\mathrm{SR}}^k\right\|_1 \tag{10}$$

$$\ell_4\left(Z, O_{\mathrm{SR}}^k\right) = \frac{1}{HW(n-1)}\sum_{i=1}^{H}\sum_{j=1}^{W}\sum_{k=1}^{n-1} \times \arccos\left(\frac{Z(i,j), O_{\mathrm{SR}}^k(i,j)}{\|Z(i,j)\|_2\left\|O_{\mathrm{SR}}^k(i,j)\right\|_2 + c}\right) \tag{11}$$

where k denotes the output feature of the kth CSOCM.

The total loss L_{total} can be expressed as

$$\mathcal{L}_{\mathrm{total}} = \partial\ell_1(Z, O_{\mathrm{SR}}) + \beta\ell_2(Z, O_{\mathrm{SR}}) + \varphi\left(\ell_3\left(Z, O_{\mathrm{SR}}^k\right) + \eta\ell_4\left(Z, O_{\mathrm{SR}}^k\right)\right) \tag{12}$$

where ∂, β, φ, and η are tradeoff factors. In the experiment, they are set to 1, 1, 0.1, and 0.1, respectively.

IV. Experiments

To demonstrate the effectiveness of the proposed CSONet, we compare our method with several image fusion methods, including traditional methods (such as 23-tap [47], GLPHS [47], and CNMF [48]) and deep learning-based methods (such as TFNET [49], SSFCNN [50], MSDCNN [15], HSRNET [27], and SSRNET [51]) on different datasets. To ensure fairness, the compared algorithms adopt the same train sets as the proposed CSONet. The implementation code of the compared algorithms is publicly available. We follow the proper parameter settings in the corresponding papers to produce the results of the compared algorithms. The proposed network is trained using PyTorch on an NVIDIA 2070 super GPU. The initial learning rate is set to 0.0005 and reduced to half after every 20 000 iterations. The model adopts the Adam optimizer.

In experiments, we use four datasets to evaluate the proposed network, which is two ground-based datasets (such as CAVE[1] and Harvard[2]) and two remote sensing image datasets (such as WDCM[3] and real data[4]). Testing the performance of the network with terrestrial hyperspectral datasets and remote sensing datasets with completely different spatial and spectral resolutions can ultimately demonstrate robustness and utility. The batch size of datasets is 32. The number of CSOCM in CSONet is set to 6.

1) *CAVE:* The CAVE dataset is a ground-based hyperspectral dataset. There are 32 hyperspectral images in the CAVE dataset; each hyperspectral image has 31 bands, covering wavelengths from 400 to 700 nm. The size of each image is 512 × 512, and the GSD is 2.5 m.
2) *Harvard:* The Harvard dataset consists of 77 hyperspectral images with size 1040 × 1392 × 31. We selected 50 indoor and outdoor scenes for training and testing. These images are captured by a commercial hyperspectral camera (Nuance FX, CRI Inc.) with the wavelength from 420 to 720 nm.
3) *WDCM:* The WDCM dataset is collected by the hyperspectral digital imagery collection experiment (HYDICE). The original WDCM dataset has a total of

[1]https://www.cs.columbia.edu/CAVE/databases/multispectral/
[2]http://vision.seas.harvard.edu/hyperspec/explore.html
[3]https://engineering.purdue.edu/ biehl/MultiSpec/hyperspectral.html
[4]https://www.l3harrisgeospatial.com/Data-Imagery/Satellite/

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TABLE I
EXPERIMENT SETTING ON DIFFERENT DATASETS

Category	Dataset	Band Number	Wavelength	Scale	Training pairs (size,number)	Testing pairs (size,number)
Ground datasets	CAVE	31	400-700nm	8	64×64×3/8×8×31,12470	512×512×3/64×64×31,12
	Harvard	31	420-720nm	8	64×64×3/8×8×31,2700	1024×1024×3/128×128×31,20
Remote sensing datasets	WDCM	191	400-2500nm	8	64×64×3/8×8×191,720	240×240×3/30×30×191,10
	Real data	8	380-1050nm	4	32×32×3/8×8×8,3871	400×400×3/100×100×8,1

210 bands. Considering that some bands are blurred by water vapor, we remove the obscured spectral bands and adopt the remaining 191 bands ranging from 400 to 2500 nm, where the size of each band is 1280 × 307.

4) *Real Data:* The real data are collected using the WorldView2 satellite. This dataset consists of pairs of real LR-MSI and HR-RGB. LR-MSI has 8 spectral bands, and the size of LR-MSI and HR-RGB is 418 × 658 and 1672 × 2632. The spatial resolution of LR-MSI and HR-RGB is 2 and 0.5 m, respectively.

A. Evaluation Metrics

In the experiment, five common evaluation metrics are introduced to evaluate the performance of the proposed method, including the peak signal-to-noise ratio (PSNR) [52], the structural similarity index (SSIM) [52], the root mean square error (RMSE) [52], the Erreur Relative Globale Adimensionnelle de Synthèse (ERGAS) [53], and SAM [54]. In particular, PSNR, SSIM, and RMSE are used to evaluate the similarity between the fused image and the reference image, and we report their average values of all spectral bands. For PSNR and SSIM, the larger the value, the better the image fusion quality. The smaller the RMSE, the better the fusion performance. The best values of SSIM and RMSE are 1, 0. ERGAS is the global mean squared error between the reconstructed hyperspectral image and the reference image. The smaller the SAM, the better the preservation of spectral information. For ERGAS and SAM, the lower the value, the better the image quality. The optimal values of ERGAS and SAM are both 0.

B. Implementation Details and Experimental Results

The experiments were conducted to evaluate the performance of a series of methods by using different datasets in both simulation and real experiments. The simulation experiment was to use the dataset as a GT image, and the corresponding LR-HS and HR-MS image pairs were obtained by spatial and spectral downsampling to evaluate the performance of the network by comparing the output image of the network with the GT image. More experimental details are shown in Table I and as follows.

1) CAVE Dataset: In the experiment, the first 20 images were selected as the training set, and the last 12 images were used as the test set. Considering the memory constraints, we do not directly feed the 20 images in the training set into the network but divide it into a series of image patches of size 64 × 64 × 31 with a stride of 16. The corresponding LR-HSI is obtained according to the Wald protocol [55], which is a common way to obtain remote sensing training sets. An 8 × 8 Gaussian blur with a standard deviation of 2 and a mean of 0 is used for the image patches and then downsampled by a factor of 8 to obtain an image patch of size 8 × 8 × 31. HR-MSI is obtained based on the spectral response of the IKONOS camera, and the size of HR-MSI is 64 × 64 × 3.

TABLE II
AVERAGE EXPERIMENTAL RESULTS OF THE PROPOSED CSONET AND OTHER FIVE METHODS ON THE 12 IMAGES OF THE CAVE TEST SET ON DIFFERENT EVALUATION METRICS. THE BOLD RED FONT DENOTES THE BEST RESULTS, AND THE BOLD BLOCK FONT DENOTES RESULT SECOND ONLY TO THE FIRST

	CAVE				
	PSNR	SAM	SSIM	RMSE	ERGAS
23-tap	25.38	3.87	0.817	0.0500	5.695
GLPHS	28.13	7.08	0.912	0.0400	3.997
CNMF	29.36	3.78	0.933	0.0528	4.855
MSDCNN	43.53	3.74	0.989	0.0078	0.896
SSRNET	43.77	3.91	0.988	0.0080	0.903
HSRNET	**46.10**	**3.21**	**0.992**	**0.0063**	**0.706**
CSONet	**48.14**	**2.36**	**0.995**	**0.0048**	**0.560**
Best value	∞	0	1	0	0

TABLE III
AVERAGE EXPERIMENTAL RESULTS ON TWO HYPERSPECTRAL IMAGES OF THE CAVE TEST SET

dataset	Method	PSNR	SAM	SSIM	RMSE	ERGAS
paints	23-tap	21.33	**7.22**	0.738	0.0845	5.507
	GLPHS	23.86	7.49	0.853	0.6204	3.987
	CNMF	30.54	**6.22**	0.937	0.0287	1.866
	MSDCNN	40.96	9.70	0.989	0.0101	0.655
	SSRNET	41.21	9.71	0.988	0.0101	0.662
	HSRNET	**42.70**	9.80	**0.991**	**0.0084**	**0.545**
	CSONet	**44.35**	9.42	**0.995**	**0.0071**	**0.461**

In Table II, we present the average experimental results of the proposed method and other state-of-the-art methods on 12 test images. Experimental results show that our proposed method outperforms other methods on all metrics, which demonstrates the powerful capability of our proposed method in hyperspectral image super-resolution. As for SAM, our proposed method outperforms the second method by 26%, which is due to the fact that our CSONet can better preserve the

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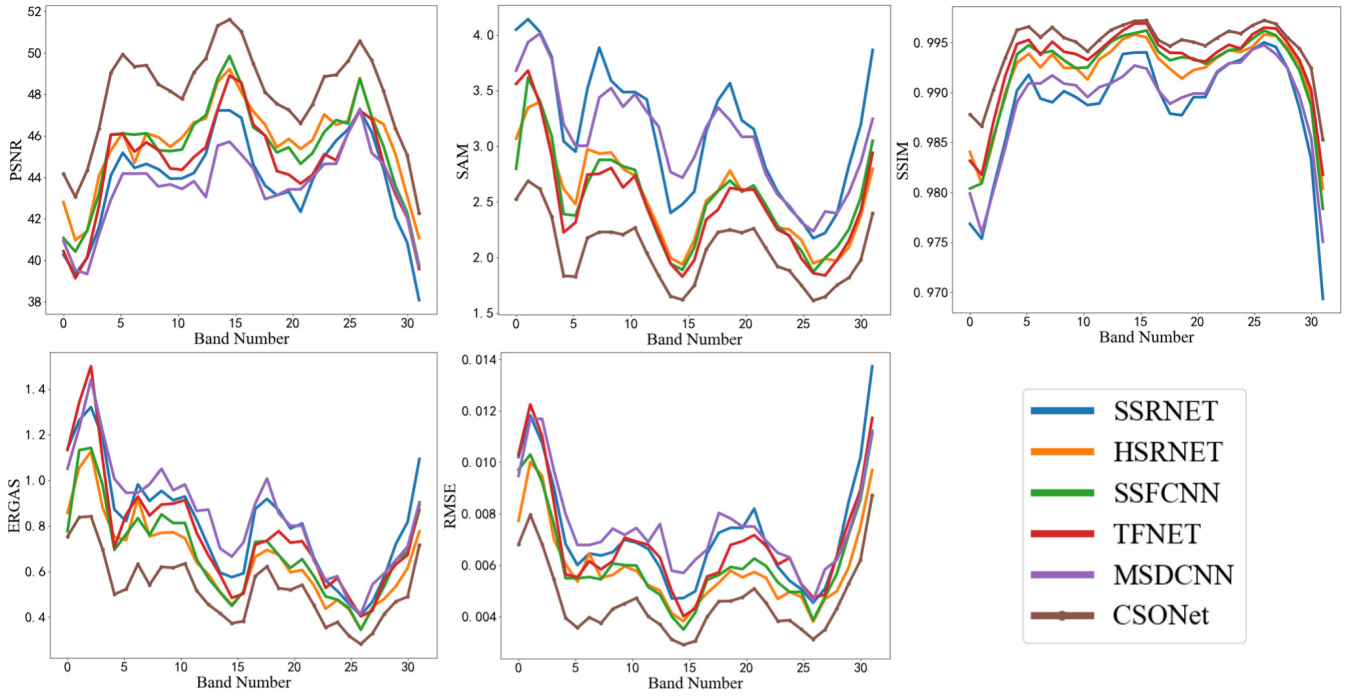


Fig. 6. Average experimental results of different methods in each band of the CAVE test set.

spectral information. Fig. 6 shows the reconstruction results of several deep learning methods in each band of the CAVE test set. It can be clearly seen that our method achieves the best experimental results in each spectral band. To visually evaluate the fusion results of these different methods, we selected an image in the CAVE test set for presentation. From Fig. 7 (left), it can be seen that the proposed CSONet can achieve better reconstruction results in both spatial texture detail recovery and spectral information preservation in comparison with the other methods. In Table III, we further show the numerical results of different methods on these images. It is obvious that the proposed method achieves the best results, which is consistent with the visual results. The above experimental results demonstrate that our proposed CSONet has strong spatial and spectral information reconstruction capabilities for hyperspectral image super-resolution.

Fig. 8 shows the learning curves of different methods on the CAVE dataset during the training. In Fig. 8, we find that our method can obtain better performance in the training phase. Fig. 9 shows comparisons of PSNR, FLOPs, and parameters. PSNRs are computed on CAVE with upscaling factor $\times 8$. We use the image from CAVE with upscaling factor $\times 8$ to compute FLOPs, where the size of HR-HSI images is $512 \times 512 \times 31$ pixels. As illustrated in Fig. 9, in comparison with some state-of-the-art methods, the proposed method can achieve better performance with a relatively large computational cost. It further shows that our proposed CSONet can obtain a good tradeoff between performance and computational cost.

2) Harvard Dataset: We selected 50 images for training and testing. The first 30 images are used for training, and the last 20 images are used for testing. The training set image patch size is $128 \times 128 \times 31$, and the division interval is 8. The processing details of the Harvard dataset are the same as the CAVE dataset.

TABLE IV
AVERAGE EXPERIMENTAL RESULTS OF DIFFERENT METHODS ON THE HARVARD TEST SET

	Harvard				
	PSNR	SAM	SSIM	RMSE	ERGAS
23-tap	33.63	23.66	0.921	0.0386	4.217
GLPHS	34.32	19.86	0.928	0.0280	3.159
CNMF	32.96	21.20	0.903	0.0323	3.617
SSRNET	46.64	3.32	0.982	0.0067	1.271
HSRNET	**48.62**	**2.93**	**0.986**	**0.0056**	**0.975**
CSONet	**48.89**	**2.84**	**0.987**	**0.0055**	**0.896**
Best value	∞	0	1	0	0

TABLE V
AVERAGE EXPERIMENTAL RESULTS OF DIFFERENT METHODS ON THE WDCM TEST SET

	WDCM				
	PSNR	SAM	SSIM	RMSE	ERGAS
23-tap	33.63	23.66	0.921	0.0386	4.217
GLPHS	34.32	19.86	0.928	0.0280	3.159
CNMF	32.96	21.20	0.903	0.0323	3.617
SSRNET	46.64	3.32	0.982	0.0067	1.271
HSRNET	**55.93**	**0.36**	**0.997**	**0.0030**	**0.066**
CSONet	**60.04**	**0.32**	**0.998**	**0.0021**	**0.047**
Best value	∞	0	1	0	0

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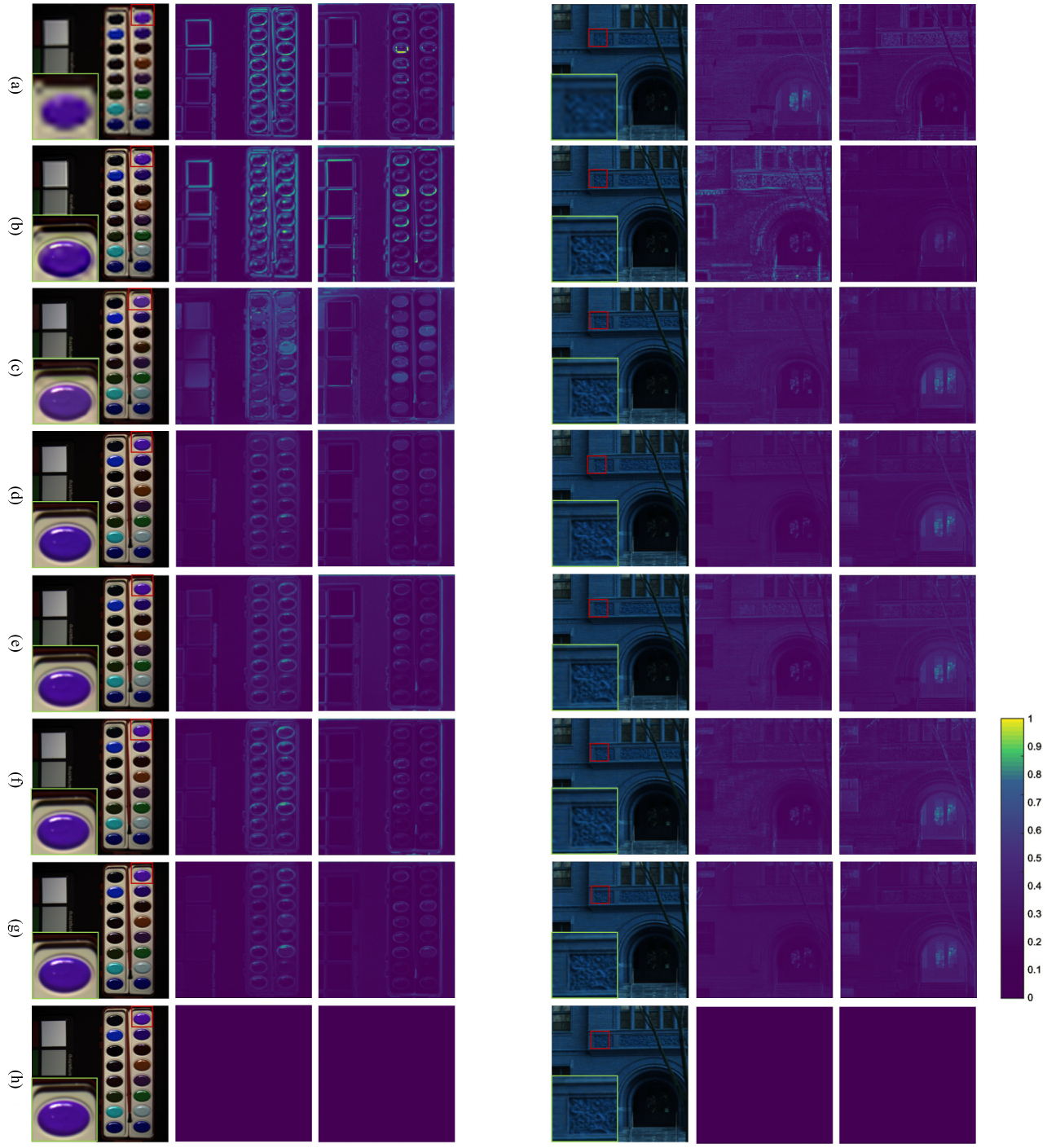


Fig. 7. Fusion results of different methods on CAVE and Harvard datasets (i.e., paints and imge6). The first and fourth columns are the pseudocolor images (R-8, G-16, B-24), the second and fifth columns are the absolute error maps between the GT images and the fused images, and the third and sixth columns are the SAM error maps between the GT images and the fused images. (a) 23-tap. (b) GLPHS. (c) CNMF. (d) MSDCNN. (e) SSRNET. (f) HSRNET. (g) CSONet. (h) GT.

Table IV shows the experimental results of different methods on the Harvard test set. As can be seen from Table IV, the experimental results of our proposed method are still the best on all metrics.

Fig. 7 (right) shows the visual comparison results. It can also be seen that our proposed method has the smallest error with the GT image.

3) WDCM Dataset: In the experiment, we take the 780 × 2632 pixel on the left-hand side of the dataset as the training set and the rest as the test set. We divide the test set with a stride of 64 and get the 10 GT images with the size of 240 × 240 in the test image patches. For the training set, we divide it into 750 image patches of size 64 × 64 with a stride of 8 as the GT images of the training pairs. The corresponding LR-HS image patches are processed in the same way as the previous datasets. The HR-MS image patches are obtained by selecting three clear bands from the 191 bands (R-20, G-55, B-182). Table V shows

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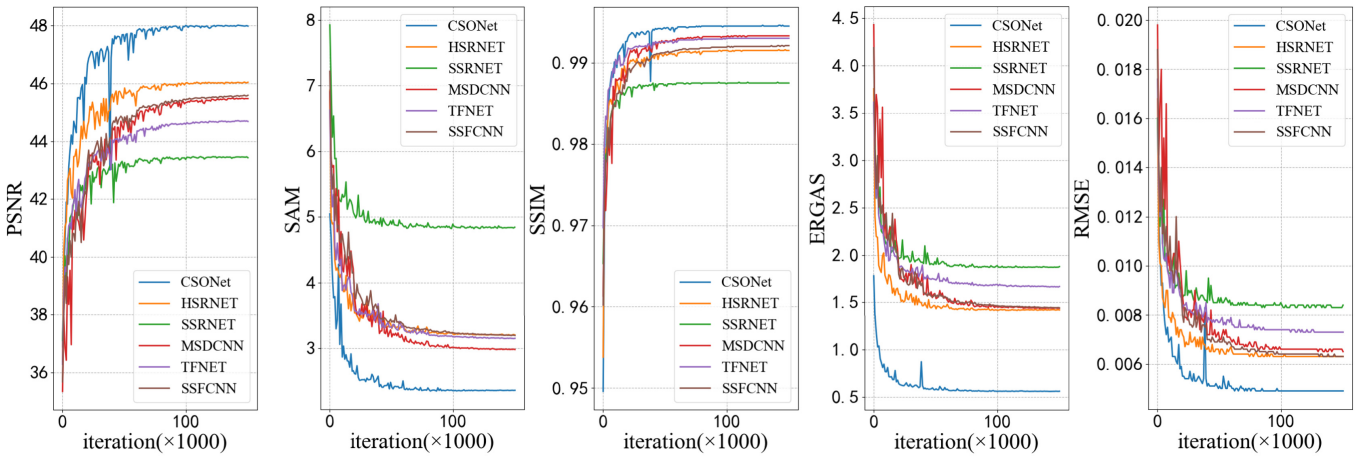


Fig. 8. Learning curves of different methods on the CAVE dataset during the training.

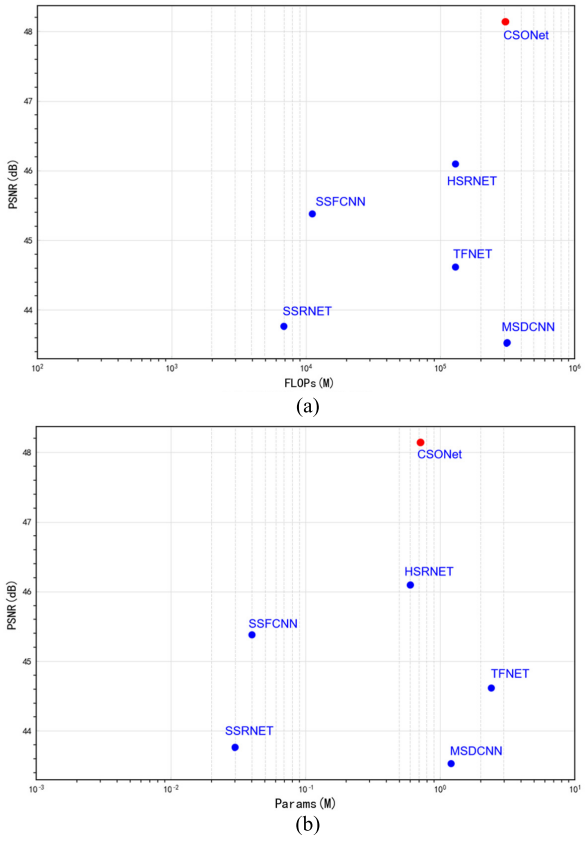


Fig. 9. Comparisons of PSNR, FLOPs, and parameters. PSNRs are computed on CAVE with upscaling factor ×8. (a) PSNR versus FLOPs. (b) PSNR versus Params.

the experimental results of different methods on the WDCM dataset.

Fig. 10 shows the visual results of different methods on the WDCM dataset. The experimental results of several traditional methods in Fig. 10 are mostly blurry, while the deep learning-based methods have shown excellent performance in recovering spatial details. Among them, SSRNET and HSRNET have large color distortions. Although MSDCNN is visually close to the GT image, it still causes some color distortion, indicating that these methods partially lose spectral information during the fusion process, and only our method achieves the best results in spatial detail and spectral information retention. As can be seen from Table V and Fig. 10, our proposed method can achieve superior experimental results in comparison with the other state-of-the-art hyperspectral super-resolution methods.

C. Ablation Experiment

In this section, we design a series of ablation experiments to demonstrate the effectiveness of each module in the proposed method.

1) Discrete Wavelet Transformation (DWT) and Deconv: Specifically, we design two variant networks to compare the necessity of downsampling with DWT, that is, $\text{CSONet}_{\text{DWT}}$ and $\text{CSONet}_{\text{conv}}$. The LR-HSI of $\text{CSONet}_{\text{DWT}}$ is directly upsampled by a factor of 8 and concatenated with the HR-MSI, instead of first upsampling the LR-HSI by a factor of 4 and then concatenated with the low-frequency information of HR-MSI extracted by DWT. The rest of the network structure remains the same. $\text{CSONet}_{\text{conv}}$ is referred to as the version that replaces the DWT module with a 3×3 convolution with a stride of 2. To demonstrate the importance of the module DWT in reconstructing spatial and spectral information, we compare three different networks: $\text{CSONet}_{\text{DWT}}$, $\text{CSONet}_{\text{conv}}$, and CSONet on the CAVE dataset. It can be seen from Table VI that, after incorporating DWT, our CSONet gives the best image fusion performance compared to these two variants. Downsampling using DWT achieves better results than convolution. The reason is probably that more efficient spatial/spectral feature information can be extracted from HR-MSI using DWT.

2) CSOCM: Considering that our network is actually similar to HSRNET, we replace the six ResNets of HSRNET with the six CSOCMs to demonstrate the effectiveness of the proposed CSOCM. The new variant network is named $\text{HSRNET}_{\text{CSOCM}}$. As shown in Table VII, after replacing the

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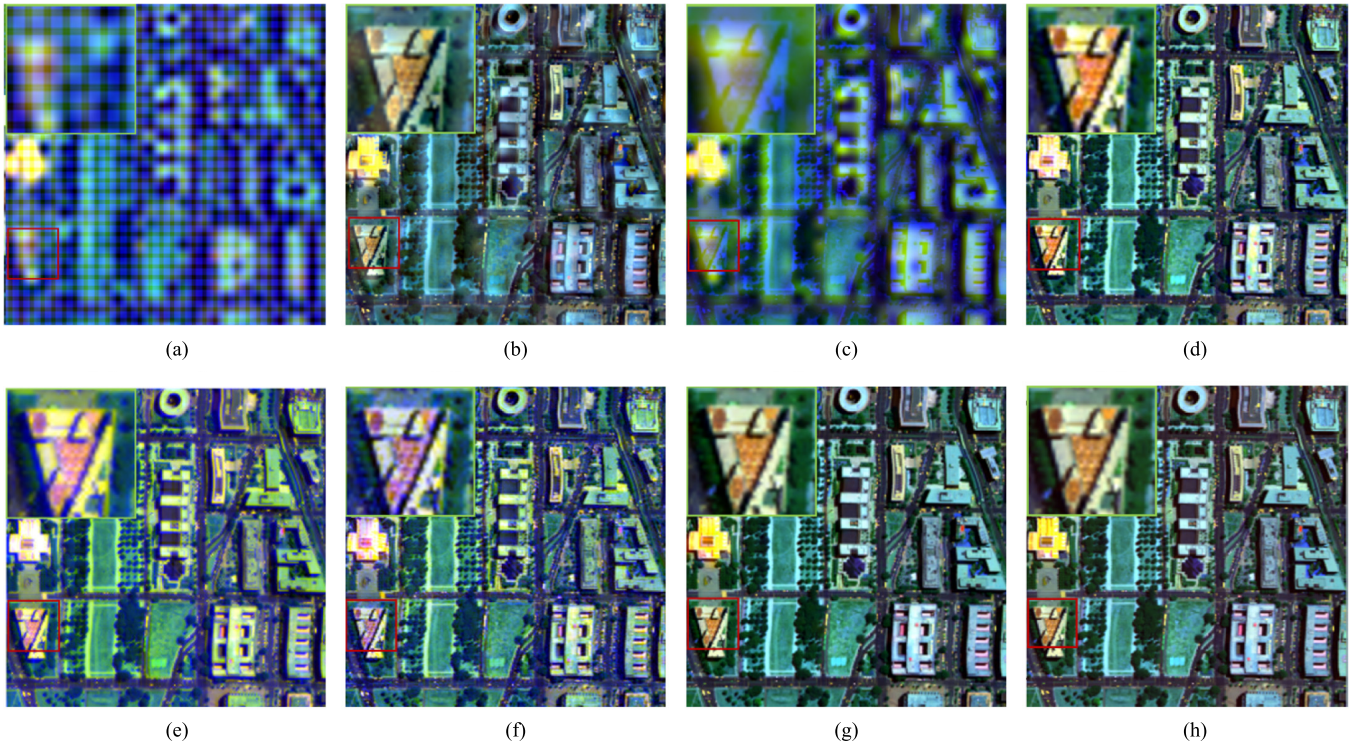


Fig. 10. Pseudocolor images of different methods on WDCM dataset (R-20, G-55, B-182). (a) 23-tap. (b) GLPHS. (c) CNMF. (d) MSDCNN. (e) SSRNET. (f) HSRNET. (g) CSONet. (h) GT.

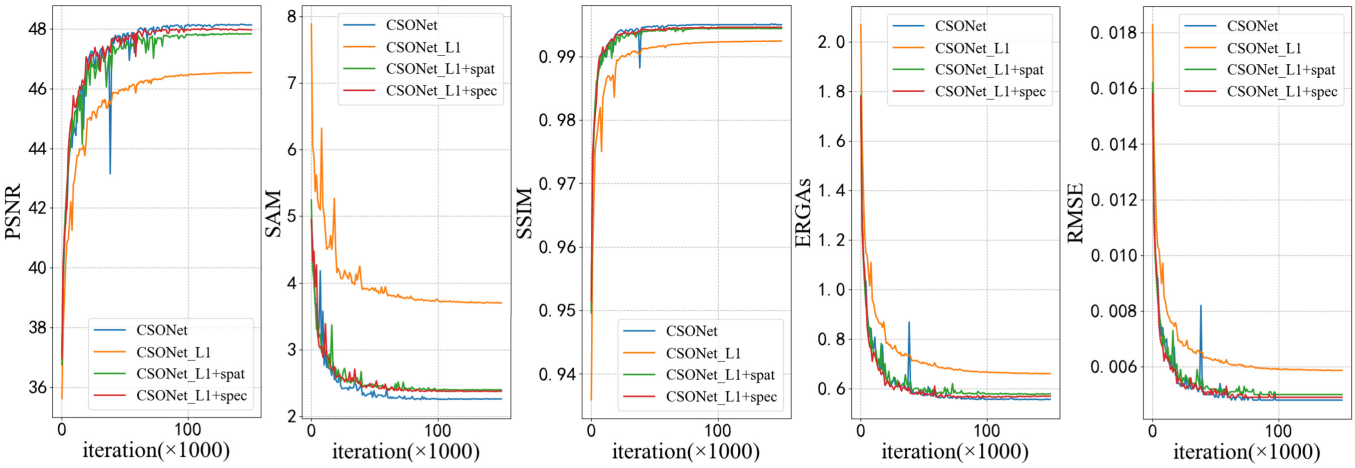


Fig. 11. Comparison of experimental results on the CAVE dataset with or without multisupervised loss functions.

TABLE VI
COMPARISON OF EXPERIMENTAL RESULTS ON CAVE DATASETS USING DWT OR CONV

	Module		Evaluation Metrics				
	DWT	conv	PSNR	SAM	SSIM	RMSE	ERGAS
CSONet$_{DWT}$			47.44	2.49	0.994	0.0053	0.595
CSONet$_{conv}$		√	**47.88**	**2.39**	**0.995**	**0.0050**	**0.595**
CSONet	√		**48.14**	**2.36**	**0.995**	**0.0048**	**0.560**

ResNet module with the CSOCM, the value of SAM is increased by 4%, and the other indicators are also improved. These demonstrate the significance of the CSOCM for image fusion.

In addition, we also conducted experiments on the number of CSOCM. The experimental results are shown in Table VIII. It can be seen from Table VIII that, when the number of CSOCMs is 6 and 7, the experiment obtains approximate results. Considering the memory consumption, we use six CSOCMs in our CSONet.

3) Multisupervised Loss: To verify the significance of multisupervised loss, we design three variant networks, which are $L1$ loss without multisupervised loss, $L1$ loss with

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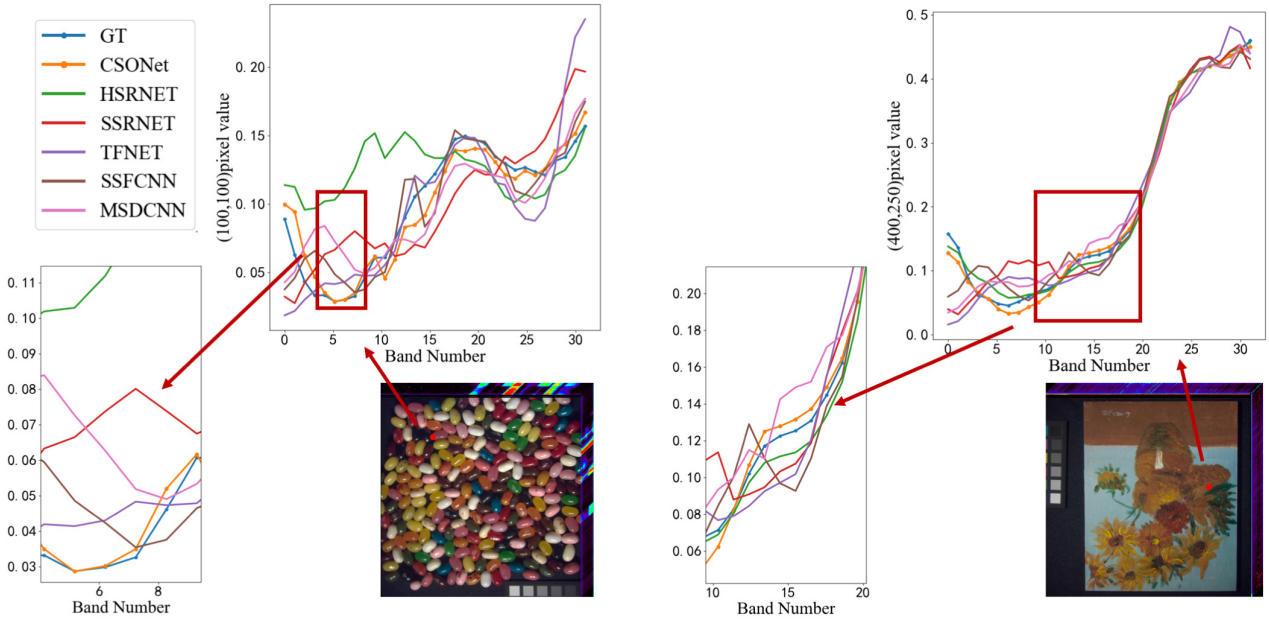


Fig. 12. Value of a pixel in each band of the two images (jelly_beans and oil_painting) by different methods.

TABLE VII

REPLACING THE RESULT OF THE HSRNET NETWORK WITH CSOCM AND THE COMPARISON OF THE EXPERIMENTAL RESULTS OF HSRNET

	CAVE				
	PSNR	SAM	SSIM	RMSE	ERGAS
HSRNET	45.69	3.30	0.992	0.0062	0.699
HSRNET$_{CSOCM}$	**46.14**	**3.17**	**0.993**	**0.0060**	**0.670**
Best value	∞	0	1	0	1

TABLE VIII

COMPARISON OF EXPERIMENTAL RESULTS OF DIFFERENT NUMBERS OF CSOCMS ON CAVE DATASETS

The number of CSOCM	CAVE				
	PSNR	SAM	SSIM	RMSE	ERGAS
3	47.74	2.42	0.994	0.0051	0.580
5	47.93	2.38	0.994	**0.0049**	0.574
6	**48.14**	**2.36**	**0.995**	**0.0048**	**0.560**
7	**48.05**	**2.36**	**0.995**	**0.0049**	**0.561**
Best value	∞	0	1	0	1

TABLE IX

COMPARISON OF EXPERIMENTAL RESULTS ON THE CAVE DATASET WITH OR WITHOUT MULTISUPERVISED LOSS FUNCTIONS

	CAVE				
	PSNR	SAM	SSIM	RMSE	ERGAS
CSONet$_{L1}$	46.54	3.08	0.992	0.0059	0.660
CSONet$_{L1+spat}$	**48.02**	**2.37**	**0.995**	**0.0049**	**0.564**
CSONet$_{L1+spec}$	47.85	2.39	0.994	0.0050	0.576
CSONet	**48.14**	**2.36**	**0.995**	**0.0048**	**0.560**
Best value	∞	0	1	0	1

multisupervised spatial loss, and $L1$ loss with multisupervised spectral loss, which are CSONet$_{L1}$, CSONet$_{L1+spat}$, and CSONet$_{L1+spec}$, respectively. From Table IX, we can find that the method without the multisupervision loss obtains relatively low accuracy. To further indicate the effectiveness of multisupervised loss, we draw the convergence curves of the above four networks and a series of deep learning methods during training in Fig. 11. The convergence process is consistent with the above analysis and shows that multisupervised loss is not only beneficial to speed up the convergence of the network but also helps to improve the model performance.

TABLE X

COMPARISON OF EXPERIMENTAL RESULTS ON THE CAVE DATASET WITH OR WITHOUT THE MSSSA MODULE

Method	CAVE				
	PSNR	SAM	SSIM	RMSE	ERGAS
CSONet$_{MSSSA}$	47.55	2.45	0.994	0.0052	0.598
CSONet	**48.14**	**2.36**	**0.995**	**0.0048**	**0.560**
Best value	∞	0	1	0	1

TABLE XI

GENERALIZATION EXPERIMENTS ON THE CAVE DATASET USING THE MODEL GENERATED FROM THE HARVARD DATASET

	CAVE				
	PSNR	SAM	SSIM	RMSE	ERGAS
TFNET	33.63	23.66	0.921	0.0386	4.217
SSFCNN	**34.32**	19.86	0.928	0.0280	3.159
MSDCNN	32.96	21.20	0.903	0.0323	3.617
SSRNET	34.27	22.74	0.916	0.0370	4.082
HSRNET	32.07	**7.02**	**0.929**	**0.0276**	**3.088**
CSONet	**37.70**	**6.41**	**0.964**	**0.0155**	**1.824**
Best value	∞	0	1	0	0

4) MSSSA Module: To demonstrate the importance of MSSSA in the proposed CSOCM, we remove the MSSSA module from the proposed network and named it CSONet$_{\text{MSSSA}}$. Table X shows the corresponding experiment results on the CAVE dataset. It can be seen that our network with the MSSSA module can obtain the ideal experiment

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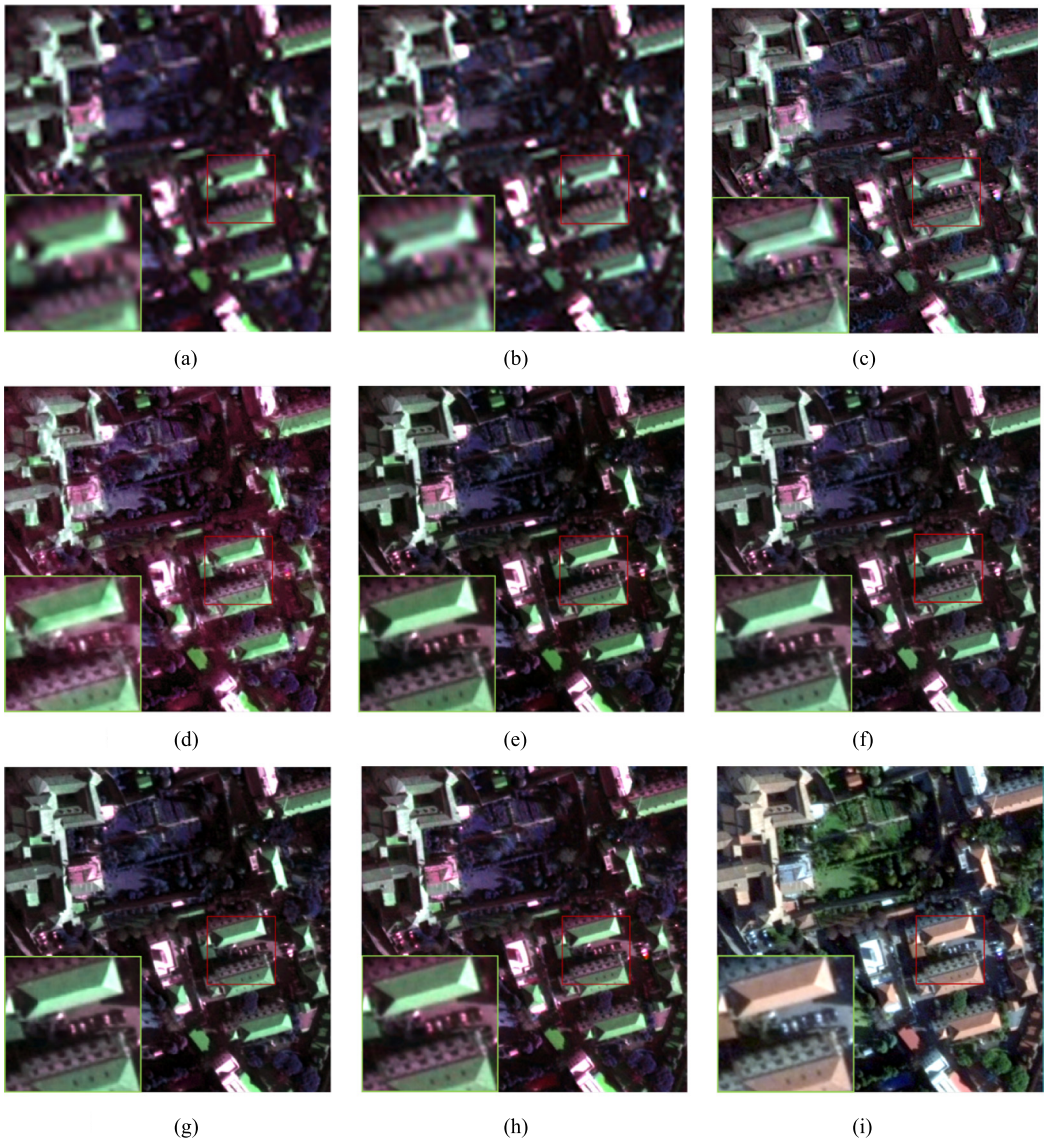


Fig. 13. Pseudocolor images of different methods on World-View2 dataset (R-2, G-5, B-3). (a) LR-HSI. (b) 23-tap. (c) GLPHS. (d) CNMF. (e) MSDCNN. (f) SSRNET. (g) HSRNET. (h) CSONet. (h) GT. (i) RGB.

results, thanks to more important spatial/spectral feature information being efficiently learned under the guidance of the proposed MSSSA.

D. Generalization Ability

In order to demonstrate the generalization ability of the proposed network, we directly test the model obtained from the Harvard training dataset on the CAVE test set. As shown in Table XI, our proposed method outperforms other methods. Moreover, Fig. 12 selects two images from the CAVE test set and shows the spectral curves of different methods and GT at a certain pixel position. This demonstrates that our method not only obtains good generation ability but also has good generalization ability.

E. Blind Super-Resolution

The above experiments have demonstrated the superiority of this method on several simulated hyperspectral datasets.

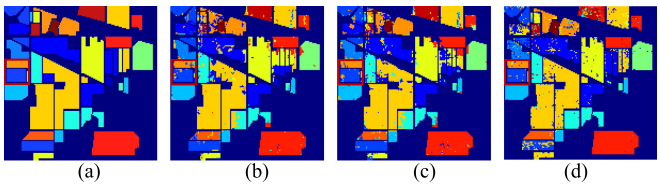


Fig. 14. Classification maps obtained by SVM over different hyperspectral reconstructed images. (a) GT. (b) HR-HSI reconstructed by CSONet. (c) HR-HSI reconstructed by $\text{CSONet}_{\text{res}}$. (d) GT IP dataset.

To further show the effectiveness of our proposed method, we have added an experiment on the blind image super-resolution at different standard deviations to show the potential of the proposed method for practical applications. In a blind super-resolution experiment, for simplicity, it is reasonable to assume that the spectral degradation matrix is known. For the spatial degenerate matrix, instead of using a fixed standard

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TABLE XII
EXPERIMENTAL RESULTS OF BLIND SUPER-RESOLUTION

value	1			1.4			1.8			2.2			2.6			3		
Evaluation Metrics	PSNR	SAM	SSIM	PSNR	SAM	SSIM	PSNR	SAM	SSIM	PSNR	SAM	SSIM	PSNR	SAM	SSIM	PSNR	SAM	SSIM
SSRNET	43.44	4.71	0.988	43.72	4.69	0.988	43.70	4.70	0.989	43.70	4.97	0.988	43.53	4.80	0.988	43.82	4.71	0.988
HSRNET	45.40	3.44	0.990	45.76	3.31	0.991	46.08	3.22	0.991	46.00	3.26	0.994	46.11	3.20	0.992	46.14	3.19	0.992
CSONet	47.59	2.43	0.994	47.90	2.39	0.994	47.96	2.39	0.994	47.77	2.43	0.994	48.13	2.35	0.995	48.15	2.35	0.995

TABLE XIII
CLASSIFICATION ACCURACY ON SVM USING DIFFERENT HYPERSPECTRAL RECONSTRUCTION IMAGES FROM IP DATASET

Method	Hyperspectral reconstruction images		
	AA	OA	Kappa
CSONet_{res}	0.772	0.849	0.827
CSONet	**0.773**	**0.854**	**0.833**
Groud Truth	**0.844**	**0.859**	**0.839**

deviation value, we assume that the blur kernel for generating the LR-HSIs is with a standard deviation in the range of [1.0, 3.0], and the Gaussian kernel is 8×8. As shown in Table XII, our CSONet achieves better hyperspectral image fusion performance.

F. Real Data

To verify the reconstruction ability of our proposed CSONet on real datasets, we conduct experiments using real data captured by the World-View2 satellite. For the test part, we crop LR-MSI and HR-RGB image patches with dimensions of 100×100 and 400×400 in the lower right corner as test image pairs, and the rest are used for training. Considering that the real dataset does not have corresponding GT images, we perform Gaussian blurring and downsampling four times on the training parts of LR-MSI and HR-RGB, respectively. Then, it is divided into 3871 pairs of $8 \times 8 \times 8$ LR-MS and $32 \times 32 \times 3$ HR-RGB images with stride 8 for training, and the corresponding original LR-MSI is used as GT image. We compare our proposed method with some methods, and the experimental results are shown in Fig. 13. It can be seen from the figure that our method has better image fusion results. This demonstrates the reconstruction ability of our proposed method on real datasets.

G. Application to Improve the Performance of Classification

The CSONet is proposed to fully extract the spatial and spectral information from LR-HSI and HR-MSI. The reconstructed hyperspectral images obtained by using the proposed CSONet can improve the accuracy of hyperspectral image classification. The specific reasons are given as follows. First, the CSOCM in CSONet is used to circulate the feature information at different scales, so as to more fully extract and fuse the features of the two images. Second, the MSSSA module enables the network to simultaneously focus on important spatial and spectral information, thereby enabling the network to incorporate more spatial and spectral information that is beneficial to reconstructed images. In order to validate the superiority of the proposed CSONet on hyperspectral image classification, we design a variant network, that is, $\text{CSONet}_{\text{res}}$, which replaces CSOCM in CSONet with ResNet, and the rest are the same as CSONet. We use the Indian Pines (IP) dataset to perform classification accuracy tests on the hyperspectral images generated by these two networks.

We employ a support vector machine (SVM) with RBF kernel for classification experiments and randomly select 25% of the samples for training. Table XIII shows the classification results on different hyperspectral images with average accuracy (AA), overall accuracy (OA), and Kappa coefficient (Kappa) as the evaluation metrics. From Table XIII, it can be seen that the classification accuracy of our proposed CSONet is better than the variant network $\text{CSONet}_{\text{res}}$ in all indicators but slightly worse than the classification accuracy of GT images. It can also be seen from Fig. 14 that our proposed CSONet has higher classification accuracy than the variant network, which proves that our proposed CSOCB has strong spatial and spectral feature extraction capabilities.

V. CONCLUSION

In this article, we propose a novel network for hyperspectral and multispectral image fusion called CSONet. It applies a progressive network, which can effectively avoid the loss of spatial and spectral information and extract rich multiscale high-resolution spatial details injecting them into the final fusion image. The proposed cross-scale octave convolution structure can learn more abundant features via cross-scale connection within a CSOCB. Besides, MSSSA is introduced to capture more informative spatial–spectral feature information at different scales. Finally, a multisupervised loss function is used to boost the flow of information and improve feature representation. Extensive experimental results demonstrate that the proposed method has better hyperspectral image reconstruction performance and strong generalization ability in comparison with recent state-of-the-art HSI super-resolution methods.

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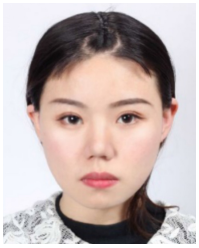
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